WaveVis inverse-sheet architecture for Moana-ocean breaking-wave linkage arrays

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AO Labs, WaveVis simulator, inverse deformation sheet record

WaveVis is an inverse simulator for a mostly flat rectangular linkage sheet with one localized plunging-wave deformation rising out of the sheet. The primary visual target is the Moana ocean reference geometry supplied for this project: a translucent, water-like sheet rising from a flat surface, swelling into a rounded crest, curling forward and downward, and returning to flat without walls, cavities, or extruded tunnel behavior. The intended artifact is not an extruded ribbon, half-pipe, swept cross-section, roof, bridge, or hollow shell. It is a full array of linkage cells over a fixed flat perimeter: the center region ramps upward, forms a rounded crest, projects a lip forward and downward, and fades smoothly back into the surrounding sheet in all directions. The reference geometry is an acceptance target, not a success claim. A generated state is still open if the human-facing render reads as a tunnel, wall-covered cavity, angular blob, missing curl, or broken linkage even when code checks pass. A verification URL is also part of the artifact: if the page ignores URL slider parameters, the screenshot is not evidence for the state Alan requested. This paper replaces the earlier narrow constraint-proof PDF as the visible system record. It defines the reference geometry, coordinate systems, generated Moana-ocean target, visible slider semantics, topology invariants, rendering views, failure modes, and verification gates required before WaveVis work can be called correct.

1 Introduction

WaveVis exists to explore whether a cellular Sarrus-style linkage array can approximate visually meaningful three-dimensional sheet deformations while preserving mechanical readability. The simulator therefore has two simultaneous jobs. First, it must generate a target shape that resembles the supplied Moana ocean references rather than a generic height field. Second, it must keep the linkage graph legible: cells, nodes, connector pairs, and edges must remain visible and connected rather than being hidden under cosmetic surfaces or erased when the shape becomes difficult.

The current design problem is stricter than drawing a pretty profile. A side silhouette can look acceptable while the actual three-dimensional sheet has a wall across the tube, a cavity where the lip should be open, an extruded-barrel cross-section, or zig-zagging linkage legs that no longer read as a mechanism. Conversely, a mesh can satisfy a local numerical check while failing the human outcome: a full rectangular sheet with a localized plunging wave. WaveVis must therefore treat the visible artifact, the mechanism topology, the parameter semantics, and the regression checks as one architecture.

The central correction captured here is that a breaking wave is not made by sweeping one two-dimensional profile sideways. A swept profile creates a tunnel, canopy, or half-pipe. The required object is a localized deformation field on a flat sheet:

$$\Phi : (x, y) \mapsto \big(x + \Delta_x(x, y),\ y + \Delta_y(x, y),\ \Delta_z(x, y)\big),$$

where the displacement is concentrated near the middle span, gradually fades toward the lateral sides, and is exactly zero on the perimeter. The curl must be strongest near the centerline and weaker toward the sides. This spatial falloff is part of the target shape, not a rendering trick.

1.1 Reference geometry

The design reference is the Moana ocean: a smooth, localized volume of water that rises out of a larger flat body, grows into a broad rounded crest, and curls into a downward lip while leaving an open underside. These reference frames are not decorative. They are the visual contract for the simulator's target geometry. A successful WaveVis state should read as a simplified linkage approximation of this water form: the outer radius is large and smooth, the inner curl radius is smaller, the lip projects forward before curling down, and the side field fades away instead of forming a constant barrel. This is

deliberately stricter than matching a side-profile curve: the full linkage array in side and isometric views must preserve the open curl without side walls, erratic zig-zags, disconnected cells, or hidden filler geometry.

Current verification boundary. This document records the target, architecture, and required gates. It does not by itself prove that the current generator has matched the supplied references. The rendered full-array artifact remains the source of truth. If the page, screenshots, or live route do not visibly match the Moana-like localized plunging wave while preserving linkage cells and flat perimeter, the correct state is “open visual mismatch,” not “reference matched.”

Figure 1. Primary supplied reference geometry. The target form is a localized plunging water sheet with a rounded crest, open underside, and forward/downward lip. The WaveVis sheet must approximate this geometry while preserving flat perimeter boundary nodes and visible linkage cells.

(a) Open tube and suspended lip.

(b) Forward/downward curl at the water edge.

(c) Localized mound emerging from flat water.

(d) Rounded column without a hard perimeter wall.

Figure 2. Additional supplied reference frames. Together they define the acceptance target: water-like surface continuity, strong center curl, smooth lateral fade, no side wall covering the underside, and no swept tunnel extrusion.

2 Target outcome

2.1 Human-facing shape

The accepted target is a “flat sheet plus localized plunging wave”:

- the outer perimeter of the rectangular sheet remains flat and fixed;
- the wave emerges through a broad, smooth ramp rather than a vertical wall;
- the crest is rounded, with no top dent;
- the lip projects forward and curls downward;
- the underside or tube is visible from side view, not covered by a side wall;
- the shape fades back to flat in every direction;
- the array remains a full linkage field with nodes and legs, not a hidden or clipped shell.

Figure 3 gives the practical acceptance boundary. The shape must read as a local wave in a sheet, not as an extrusion. A single cross-section can guide the profile, but the generated three-dimensional field must vary across span.

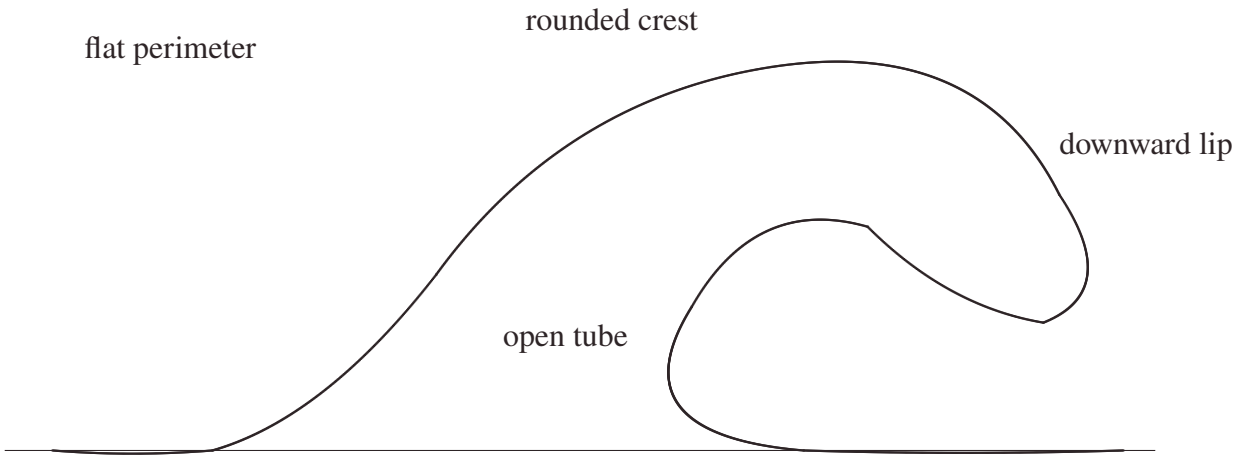

Figure 3. Desired side silhouette as an outcome object. The drawing is not a literal mesh recipe. It is the visible side-view contract: smooth back ramp, larger outer radius, smaller inner radius, curled lip, and flat return. The three-dimensional solver must then localize this silhouette in the middle of a full sheet rather than sweep it as a constant tunnel.

2.2 Hard constraints

The WaveVis generator and renderer must preserve the following hard constraints:

1. **Flat perimeter.** Boundary nodes must remain on the rest sheet. No side, front, or rear perimeter should be pulled into the wave.

2. **Full array visibility.** The rendered artifact must show the actual linkage grid. Hiding or ghosting broken cells does not count as a fix.

3. **Connector closure.** A cell connector is an actual shared graph location. Lines should meet at nodes; apparent detached legs are a failure.

4. **No erratic legs.** Long arms, spikes, zig-zags, random backtracking, and visible overlaps indicate a broken geometry or render path.

5. **Smooth strain sharing.** Adjacent rows and columns should change gradually enough that the mesh reads as a continuous mechanism field.

6. **Localized wave.** The curl must be strongest near the center and fade laterally. A constant side-to-side tunnel is not the target.

7. **Side/cross-section separation.** Side view shows the side view of the full three-dimensional render. Cross-section view is the only mode that intentionally slices or projects the center profile.

8. **No filler geometry.** Cosmetic walls, caps, planks, bridges, hidden couplers, or masks must not be used to conceal a bad mechanism.

3 Coordinate model

The simulator starts from a rectangular rest lattice. A node with row i and column j has rest position

$$p_{ij}^0 = (x_j, y_i, 0).$$

The target is a displacement field

$$p_{ij}^* = p_{ij}^0 + m\, D(x_j, y_i; \theta),$$

where $m \in [0, 1]$ is the morph amount and θ represents all shape parameters except rigid placement. The morph parameter only interpolates between rest and the finalized target. It must not change the target shape definition, remesh the lattice, or introduce a different topology.

The longitudinal coordinate $u \in [0, 1]$ measures progress through the wave region from rear/root to front/lip. The span coordinate $v \in [-1, 1]$ measures lateral distance from the centerline. A good WaveVis target uses both:

$$D(x, y) = A(u, v)\, P(u) + B(u, v)\, Q(u),$$

where $P(u)$ is the side-profile contribution, $Q(u)$ can include forward curl or secondary displacement terms, and A, B are smooth two-dimensional envelopes that go to zero at the sheet perimeter.

The key architectural rule is that $A(u, v)$ is not constant in v. A constant span envelope turns any attractive profile into a tunnel. A localized plunging wave requires $A(u, 0)$ large near the centerline and $A(u, \pm 1) \approx 0$ near the sides, with a monotone or smoothly unimodal falloff rather than a hard wall.

4 Generation pipeline

The target-building pipeline is:

1. sanitize rows, columns, visible sliders, hidden legacy shape parameters, display flags, and URL parameters;
2. build the full rectangular node grid;
3. compute the canonical Moana-ocean wave displacement from the generated target;
4. apply the span envelope that localizes the wave in y;
5. preserve the boundary by forcing perimeter displacement to zero;
6. apply morph interpolation from rest to target;
7. apply rigid steer and position transforms after the shape is finalized;
8. build full rectangular edge and quad topology;
9. compute strain, bend, shear, dihedral, and display metrics;
10. render the full array, side view, or cross-section without substituting fake surfaces.

The distinction between the generated target and rigid placement parameters is essential. The visible app no longer exposes manual xz or xy profile editors. The generator owns the Moana-ocean target. The user-facing sliders are intentionally narrow: scale uniformly resizes that target, morph blends from rest to the completed target, steer rotates the finished target around the anchor, and position translates the finished target horizontally. Neither steer nor position may feed back into profile curvature, curl, width, strain generation, or cell topology.

Stage	Contract	Failure caught
Target profile	Defines the desired side silhouette.	Blob, neck/head, hill without lip, top dent.
Span envelope	Localizes the wave in the sheet.	Swept barrel, side wall, U-shaped tube blocker.
Boundary clamp	Keeps the perimeter flat.	Pulled sheet edges, non-flat perimeter.
Morph	Interpolates rest to target only.	Half-morph mangling, topology change while sliding.
Topology	Keeps full node/edge/quad graph.	Missing grid, hidden cells, detached legs.
Renderer	Shows the actual mechanism.	Fake shell, hidden wall, side view as cross-section.

Table 1. Pipeline stages and the failure modes each stage must prevent.

5 Generated Moana target

The current visible app does not ask the user to draw a profile. That path created too much ambiguity and repeatedly made the simulator look like a swept tunnel or hand-edited cavity rather than a single coherent wave. The source may still contain historical helper functions for compatibility, but the user-facing target is generated: a localized Moana-ocean deformation with a broad rear ramp, rounded crest, forward-and-down lip, open underside, smooth lateral fade, and flat rectangular perimeter.

The generated target is a reference field, not a set of cell locations. The final cell grid is determined by row and column counts, then sampled against that target. A clean state must keep the full linkage array visible while making the centerline and the lateral silhouette read as one plunging wave. The generator must never stitch the lip directly to the ground or draw a hard polygon boundary around a former editor curve, because that creates the wall, bowl, and cavity failures this record explicitly rejects. The curl is also resolution-sensitive: under-resolved column counts are promoted to the smooth-wave minimum because honoring a compact column request recreated the visible dimple and made the side silhouette fail.

6 Slider semantics

Control	Intended effect	Must not do
`morph`	Blend rest sheet into the completed target.	Remesh, flip cells, create zig-zags, change profile identity.
`scale`	Uniformly enlarge or reduce the generated Moana-ocean target.	Change steer, position, topology, or the target's wave identity.
`steer`	Rigid yaw rotation of the finished wave.	Bend, stretch, or reshape the wave.
`position`	Rigid horizontal translation of the finished wave.	Change silhouette, strain field, or curl.
display toggles	Surface, flat grid, cells, and connectors are display layers only.	Hide broken geometry or substitute a fake repair path.

Table 2. Visible control contract after removing manual profile editors and hidden shape sliders. The generated Moana-ocean target is the default shape; visible controls must preserve that shape unless their narrow role explicitly changes scale, morph, steer, position, or display layers.

7 Breaking-wave geometry

The desired generated-mode profile is closer to a localized plunging wave than to a cone, cylinder, or arch. The useful approximation is a tapered, asymmetric surface whose highest crest sits behind the curl, whose outer radius is larger than the inner radius, and whose underside remains open. Along the centerline, a breaking wave profile can be represented as three joined curve families:

$$P(u) = \begin{cases} P_{\text{ramp}}(u), & 0 \le u < u_c, \\ P_{\text{crest}}(u), & u_c \le u < u_l, \\ P_{\text{lip}}(u), & u_l \le u \le 1. \end{cases}$$

The ramp rises from the sheet. The crest rounds over. The lip turns forward and downward. The derivative at the terminal lip should point downward when `lipDip` $> 90°$:

$$\left.\frac{dz}{dx}\right|_{\text{tip}} < 0.$$

The final point is allowed to be lower than the farthest horizontal reach. In a real breaking wave, the maximum reach can occur at the shoulder while the lip tip turns down from that shoulder. The downturn must remain open and forward-readable from the side; it must not fold into a closed pucker, dimple, bowl, or side-wall pocket. Treating the farthest-right point as the tip is the mistake that

produces a straight falling chute, while overcorrecting into an under-tucked return is the mistake that produces the rejected cavity. The underside return must also advance back into the flat sheet after the lowest lip sample; a backward floor point is disallowed because it reads as a small lower-lip dimple even when it does not trip a larger bowl metric.

The three-dimensional field must multiply this centerline motion by a span falloff:

$$E(v;u) = \exp\left[-\left(\frac{|v|}{w(u)}\right)^{\alpha}\right],$$

with $w(u)$ varying through the wave. The curl should have a smaller active width than the broad back ramp, so the lip is localized and the tube remains visible from the side.

8 Mechanism topology

WaveVis uses a rectangular graph of nodes, horizontal profile edges, vertical span edges, and quads. Each interior node participates in the lattice; each cell is bordered by edges; and each connector must visually close. The graph is not allowed to become a set of isolated rows.

The mechanism evidence from the earlier proof remains relevant. The all-four-equal X-cell branch was rejected because the finite connector assignment does not close cleanly under the current overhang contract. The accepted branch is pairwise: east-west legs form one equal collinear pair, north-south legs form another equal collinear pair, and the angle between the pairs is allowed to vary. This pairwise rule preserves a readable linkage without forcing impossible equal-arm closure (*1*).

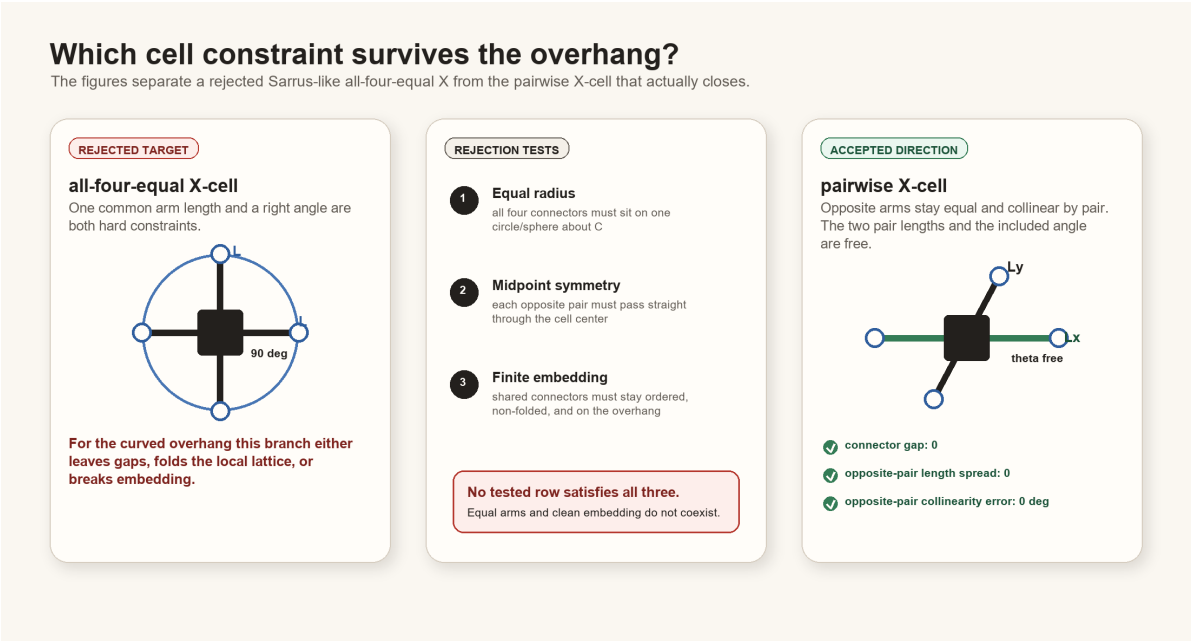

Figure 4. Mechanism contract inherited from the constraint proof. WaveVis should preserve exact shared contact and pairwise equal, collinear opposite arms rather than hiding infeasible all-four-equal geometry behind filler surfaces.

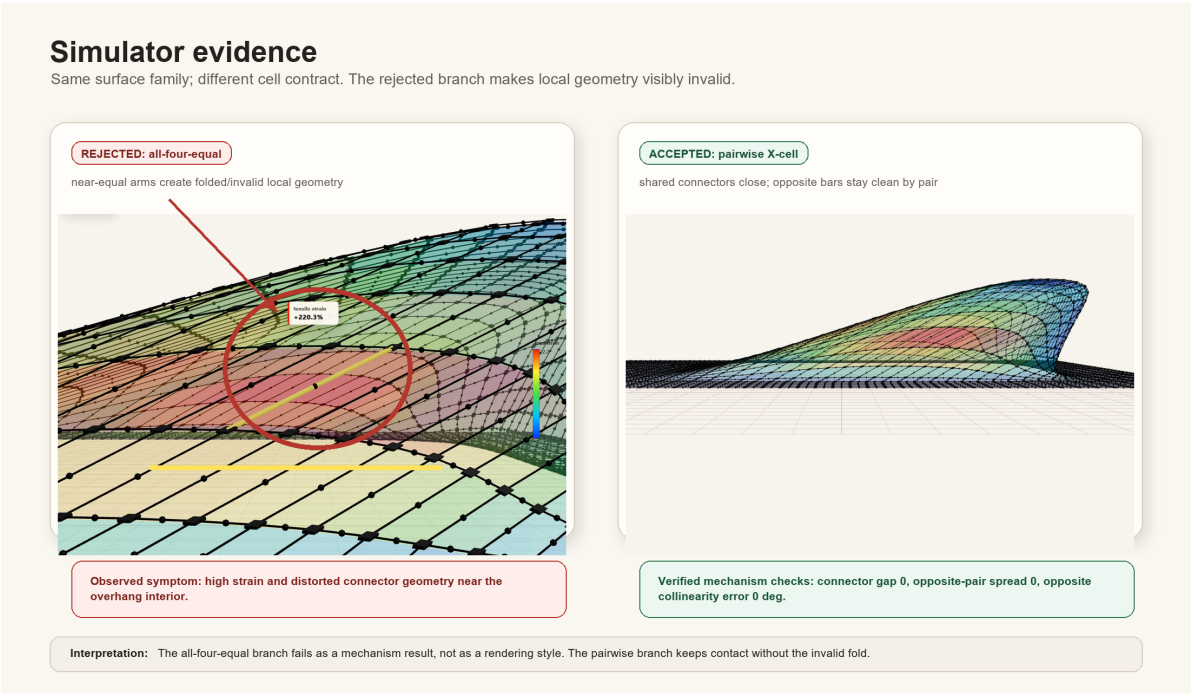

Figure 5. Simulator evidence for the linkage direction. The current architecture must keep this mechanism truth visible while the inverse target shape is improved.

9 Rendering modes

WaveVis has at least three distinct views:

- **3D/isometric view** shows the full rectangular array in perspective.
- **Side view** is a camera view of the full three-dimensional render from the side. It is not a cross-section and must not slice away rows.
- **Cross-section view** intentionally shows a selected slice or profile. This is the only mode that may look like a profile projection.

The view contract matters because a side view can hide the curl if lateral geometry forms a wall over the tube. In that case the correct fix is to reshape the three-dimensional surface so the lateral envelope follows the wave contour and fades out, not to switch the side view into a cross-section or hide the wall.

10 Verification gates

The generator should be checked with both numeric and visual gates. Numeric checks are not sufficient, but they catch repeated hard failures before visual review.

Gate	Measurement	Acceptance intent
Perimeter flatness	$\max_{\partial\Omega} \|p^* - p^0\|$.	Boundary displacement is zero or visually negligible.
Topology completeness	Expected node, edge, and quad counts.	Full rectangular sheet graph remains visible.
Connector closure	Edge endpoints meet their node coordinates.	No detached legs or fake bridges.
Lip curl	Crest height minus lip-tip height; terminal slope.	The lip curls down rather than ending as a hill.
Span localization	Row-wise height profile across y.	Center strong, sides faded; no swept tunnel.
Morph continuity	Residual between adjacent morph samples.	No sudden jumps or topology flips.
Slider isolation	Aligned geometries across position/steer/width changes.	Rigid controls stay rigid; width stays span-only.
Visual side check	Browser side view screenshot or direct render inspection.	Curl visible without cross-section mode.
Defect repair loop	Confirmed visible defects re-enter source repair.	Acknowledging dimples, pockets, walls, or missing curl is not a stopping state.
`noVisibleDimplePocket`	Open downturn, return advancement, and curl clearance in the centerline.	The tip may curl downward, but it must not form a tight pucker, pocket, dimple, or enclosed bowl.
`startupUrlParamCoverage`	Startup parser covers all public and legacy slider aliases.	A shared verification URL cannot silently load a different height, overhang, lip dip, width, or smoothing state.

Table 3. Verification gates. A build passing TypeScript is not enough; WaveVis must pass the rendered outcome Alan actually uses.

11 Failure modes

The most important failure classes are now explicit:

1. **Swept barrel.** One side profile is dragged sideways at constant cross-section. The result reads as a tunnel, roof, half-pipe, or bridge.

2. **Side-wall cover.** The lip may exist in a cross-section, but full side view shows a wall covering the tube.

3. **Straight chute.** The crest drops nearly vertically to a flat shelf instead of curling forward and under.

4. **Blob or lump.** Excess smoothing or broad support creates a mound with no wave identity.
5. **Top dent.** The outer crest has a kink or concave dent where a smooth radius is required.
6. **Dimple, cavity, or bowl.** The surface creates a pocket, pucker, enclosed hollow, or bowl-like depression rather than a plunging sheet.
7. **Stale URL state.** A browser URL contains height, overhang, width, lipDip, or smoothing values, but startup ignores them and renders a different state.
8. **Zig-zag linkage.** Neighboring legs alternate direction or length erratically.
9. **Hidden mechanism.** Broken topology is ghosted, clipped, or replaced by a cosmetic surface.
10. **Passive defect acknowledgement.** A visible problem is named in chat or recorded in this paper, but the generator and rendered artifact are not repaired and rechecked.

The solution direction is to simplify the target field while strengthening the gates: localized smooth wave first, full array second, mechanism-clean render third. Complexity should be added only if it improves one of those outcomes without breaking the others.

12 Materials and Methods

The implementation source is the WaveVis React and TypeScript app (*2*). The main geometry source is `src/latticeGeometry.ts`. The primary render source is `src/LatticeViewer3D.tsx`. Controls are declared in `src/TargetShapeControls.tsx`. Regression checks are run through `scripts/check-inverse-sheet.cjs`.

The public app serves a static build. The architecture PDF is placed in the public proofs directory as `wavevis-system-architecture.pdf` and linked from the WaveVis control headers. The old connector proof remains a source artifact for the mechanism decision, but the visible paper action now points to this architecture record because the current work is broader than connector assignment.

13 Discussion

The main design lesson is that WaveVis cannot be corrected by tuning only the side profile. The user-facing failure was three-dimensional: side walls, swept tubes, missing full-array linkages,

broken side/cross-section semantics, and zig-zag legs. The architecture therefore now uses a single generated target contract rather than a manual editor contract. The sheet generator is responsible for creating a localized Moana-like displacement field with lateral fade, boundary preservation, topology completeness, and a visible curled lip.

This distinction also clarifies future work. If the simulator produces a good cross-section but a bad side view, the cross-section is only evidence that the profile exists somewhere. The real artifact still fails if the full three-dimensional render hides the curl or reads as an extrusion. Similarly, if the mesh is clean but the wave looks like a low mound, the mechanism is not enough. Both the shape and the linkage must be correct.

A newer lesson is that defect naming is not progress by itself. If a rendered state shows dimples, pockets, side walls, or a missing curl, the correct workflow is source repair, geometry-check update, rendered screenshot verification, and public-bundle verification. The current check names this directly as `noVisibleDimplePocket`: the lip must remain an open downturned branch with visible clearance and a return that advances into the flat sheet instead of folding into a pucker. The same source-of-truth rule applies to URL state. Several correction screenshots used URLs that carried explicit height, overhang, width, lipDip, and smoothing values; ignoring those parameters caused the page to render a different geometry than the requested test state. The startup parser is now part of validation through `startupUrlParamCoverage`. The architecture record is useful only if it changes the generator, state loader, and acceptance loop; it cannot substitute for a corrected human-facing WaveVis render.

14 Acknowledgements

This record was written to reduce repeated handoff burden during WaveVis development. The repeated complaints about side walls, missing cells, zig-zag legs, cross-section confusion, and swept-barrel geometry are treated as acceptance criteria rather than optional polish notes.

15 Funding

No external funding is claimed in this system architecture record.

16 Author contributions

Alan N. Pham defined the target artifact, accepted and rejected visual states, and the mechanism constraints. Codex generated this architecture record from the current WaveVis source and the active development contract.

17 Competing interests

The author declares no competing interests for this internal architecture record.

18 Data availability

The source data for this record is the WaveVis source tree, the Alan-supplied Moana ocean reference images copied into the proof figure set, the local proof diagrams, and the current simulator behavior. The public PDF is served with the WaveVis static app.

19 Code availability

The relevant implementation files are `src/latticeGeometry.ts`, `src/LatticeViewer3D.tsx`, `src/TargetShapeControls.tsx`, and `scripts/check-inverse-sheet.cjs` in the WaveVis app repository.

20 Additional information

This document supersedes the old header link to the narrow connector-contact constraint proof as the primary public WaveVis PDF. The connector proof remains an implementation reference for mechanism feasibility, but the current public artifact needs the broader system architecture and outcome-verification contract.

21 References

1. A. N. Pham, *WaveVis overhang connector assignment: evidence rejecting all-four-equal X-cells*, Internal WaveVis constraint proof and simulator evidence figures, 2026.

2. A. N. Pham, *WaveVis inverse-sheet simulator source tree*, Local source files: src/latticeGeometry.ts, src/LatticeViewer3D.tsx, src/TargetShapeControls.tsx, scripts/check-inverse-sheet.cjs, 2026.